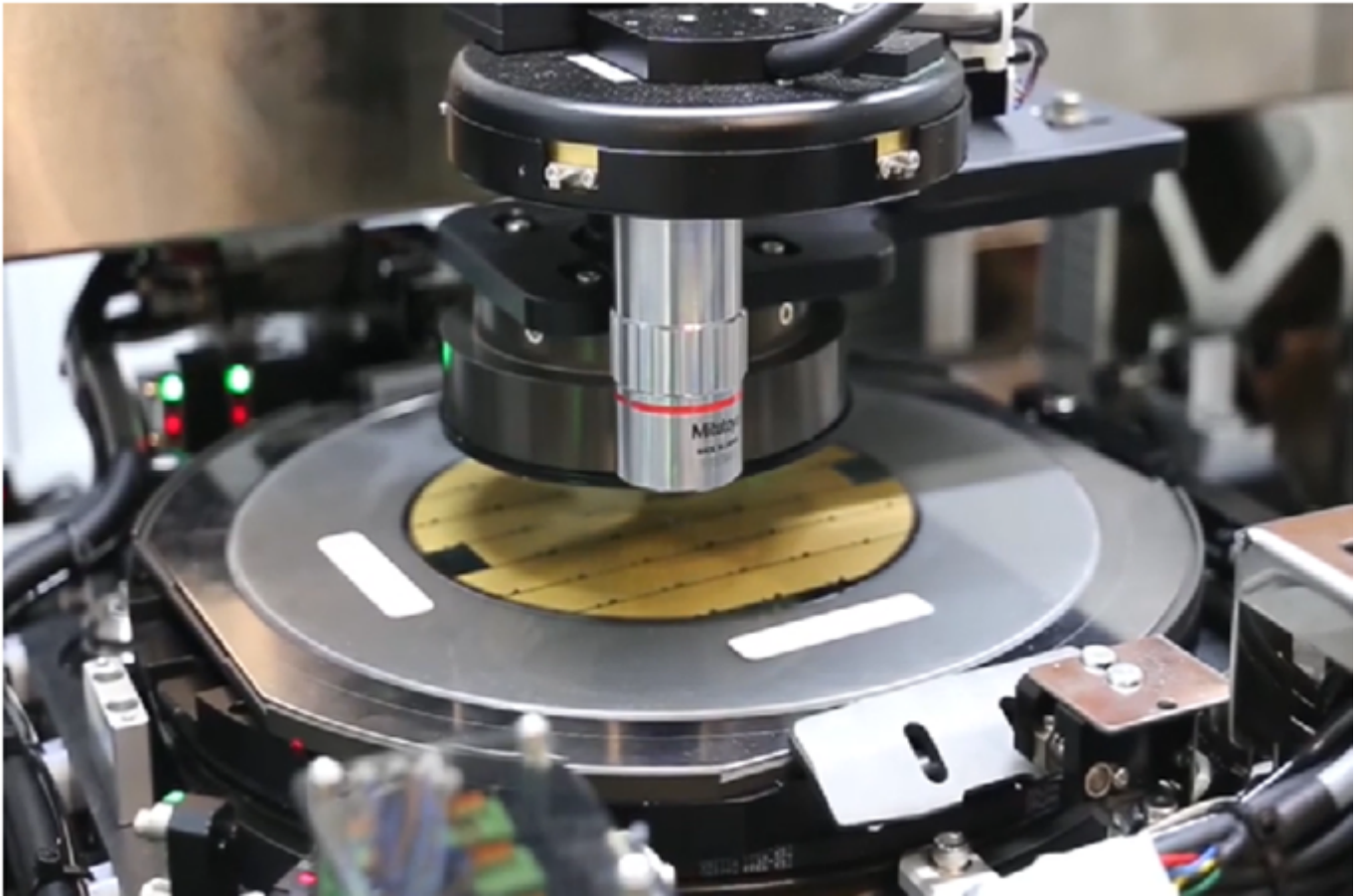


Qualification Pack



Assembly Process Technician - Wafer Testing

QP Code: TEL/Q7201

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

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TEL/Q7201: Assembly Process Technician - Wafer Testing

Brief Job Description

The individual for this job role is responsible for performing various tasks associated with the testing and quality control of semiconductor wafers used in telecom applications. They are also responsible to work with sophisticated testing equipment and follow established procedures to ensure the integrity and performance of the wafers. In addition, the individual also document test results, identify any defects or issues, and collaborate with engineering teams to implement corrective actions. The individual for this job role requires a keen attention to detail and a strong commitment to maintaining high standards of quality.

Personal Attributes

The individual must demonstrate attention to detail, analytical and problem-solving abilities, and effective communication skills, both written and verbal. The individual should be adept at working collaboratively with others.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N7201: Assemble Wafer Test Equipment](#)
2. [TEL/N7202: Perform Wafer Test Procedures](#)
3. [TEL/N7203: Analyze Wafer Test Data](#)
4. [TEL/N7204: Maintain Wafer Test Equipment](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Semiconductor-Manufacturing & Packaging
Occupation	Semiconductor-M&P
Country	India
NSQF Level	4.5
Credits	15

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/3522.9900
Minimum Educational Qualification & Experience	Graduate (completed 1st year of 3- years/4 years UG) with NA of experience OR 12th grade Pass (Electronic/Telecom/IT) with 1.5 years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	28/02/2026
NSQC Approval Date	27/08/2024
Version	1.0
Reference code on NQR	QG-4.5-TL-02953-2024-V1-TSSC
NQR Version	1.0

Qualification Pack

TEL/N7201: Assemble Wafer Test Equipment

Description

This NOS unit focuses on preparing the wafer test environment, setting up equipment, and loading wafers for testing as per defined procedures.

Scope

The scope covers the following :

- Prepare Test Environment according to SOPs (Standard Operating Procedures)
- Set up & Calibrate Wafer Test Equipment
- Load Wafers into the Test Equipment Following Handling Protocols

Elements and Performance Criteria

Prepare Test Environment according to SOPs

To be competent, the user/individual on the job must be able to:

- PC1.** review SOPs (Standard Operating Procedures) for specific test requirements.
- PC2.** identify and gather all necessary materials and tools listed in the SOPs for wafer testing
- PC3.** verify the functionality and proper calibration of any required measuring tools.
- PC4.** consult equipment recommendations or SOPs for the ideal temperature and humidity range for optimal test performance
- PC5.** utilize thermometers and hygrometers to monitor the test environment conditions
- PC6.** adjust environmental controls (e.g., air conditioning, dehumidifiers) as needed to maintain the specified temperature and humidity levels.
- PC7.** verify that essential utilities like compressed air and power supply are available and functioning properly
- PC8.** conduct a visual inspection of utility lines and connections for any signs of damage or leaks
- PC9.** test the functionality of compressed air pressure and power supply voltage according to established procedures
- PC10.** document the test environment conditions (temperature, humidity, etc.) and any adjustments made according to SOPs

Set up & Calibrate Wafer Test Equipment

To be competent, the user/individual on the job must be able to:

- PC11.** analyze the type of telecom wafers to be tested
- PC12.** review test program requirements for specific functionalities to be evaluated
- PC13.** identify the appropriate test equipment model from available options based on wafer type & test needs and consult SOPs for the selected test equipment model
- PC14.** identify and gather required test accessories like test heads and probes suitable for wafer testing
- PC15.** install the accessories onto the designated ports or fixtures on the test equipment, following established procedures, ensuring proper connection and secure positioning of the test heads and probes for optimal test performance

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- PC16.** configure relevant equipment parameters (e.g., voltage levels, signal frequencies), based on the specific test program
- PC17.** ensure parameter settings align with the requirements outlined in the test program for accurate testing of functionalities (e.g., communication channel bandwidth, data integrity)
- PC18.** conduct a visual inspection of the test heads and probes to ensure proper alignment and positioning for contacting the designated test points on the wafers
- PC19.** utilize calibration tools or procedures outlined in SOPs to verify the calibration of test heads and probes
- PC20.** perform necessary adjustments or report the issue for maintenance personnel to address the issue, if calibration is out of specification
- PC21.** initiate a system check function on the test equipment as per SOPs
- PC22.** verify that all components of the test equipment, including accessories, are functioning properly as a system before proceeding with actual testing
- PC23.** identify the relevant test parameters requiring calibration for the wafer test
- PC24.** perform calibration routines for each identified parameter according to established procedures
- PC25.** document calibration results, including dates, performed adjustments, and reference standards used, for traceability purposes

Load Wafers into the Test Equipment Following Handling Protocols

To be competent, the user/individual on the job must be able to:

- PC26.** review SOPs for specific handling procedures for the type of wafer being tested
- PC27.** visually inspect the wafers for any physical damage or defects that might affect testing
- PC28.** clearly label the wafers with identification information according to established protocols for traceability throughout the testing process
- PC29.** select and utilize appropriate handling tools (e.g., vacuum tweezers, transport carriers) designed for safe handling of delicate wafers
- PC30.** refer to SOPs or the test equipment manual for the designated loading pattern specific to the type of wafer being tested
- PC31.** verify that all wafers are loaded according to the pattern to optimize testing efficiency and avoid potential issues during the testing process
- PC32.** double-check the orientation of each loaded wafer to ensure it aligns with the test head and probe configuration for proper contact with the test points
- PC33.** verify that all wafers are securely positioned within the carrier or fixture to prevent any movement or misalignment during testing
- PC34.** secure the wafer carrier or fixture within the test equipment using the designated locking mechanisms
- PC35.** implement established electrostatic discharge (ESD) control procedures throughout the wafer handling process
- PC36.** utilize ESD-safe tools (e.g, grounded tweezers), wear ESD wrist straps grounded to the equipment, and maintain a clean and ESD-safe work environment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** importance of maintaining a clean and controlled environment for wafer testing
- KU2.** principles of contamination control in a semiconductor cleanroom environment
- KU3.** proper handling and storage procedures for test materials and tools
- KU4.** environmental factors (e.g., temperature, humidity) affecting test equipment performance
- KU5.** principles of operation for different types of wafer test equipment
- KU6.** functions and proper use of various test equipment accessories (e.g., test heads, probes)
- KU7.** interpretation of test program specifications and configuration parameters
- KU8.** importance of proper alignment and calibration of test heads and probes for accurate testing
- KU9.** significance of regular calibration for maintaining test equipment accuracy
- KU10.** different types of calibration procedures for various test parameters
- KU11.** use of calibration standards and traceability requirements
- KU12.** documentation procedures for calibration results and maintenance records
- KU13.** safe handling techniques to prevent damage to delicate wafers (e.g., electrostatic discharge control)
- KU14.** importance of proper labeling and identification of wafers for traceability
- KU15.** designated loading patterns and procedures for specific test equipment models
- KU16.** visual inspection techniques to identify potential defects on wafers before testing

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** listen attentively to understand the information/ instructions being shared
- GS4.** communicate politely and professionally
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** coordinate with the co-workers to achieve the work objectives
- GS7.** evaluate all possible solutions to a problem to select the best one
- GS8.** take quick decisions to deal with workplace emergencies/ accidents

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare Test Environment according to SOPs</i>	10	14	-	3
PC1. review SOPs (Standard Operating Procedures) for specific test requirements.	1	1	-	-
PC2. identify and gather all necessary materials and tools listed in the SOPs for wafer testing	1	2	-	1
PC3. verify the functionality and proper calibration of any required measuring tools.	1	2	-	1
PC4. consult equipment recommendations or SOPs for the ideal temperature and humidity range for optimal test performance	1	1	-	-
PC5. utilize thermometers and hygrometers to monitor the test environment conditions	1	1	-	-
PC6. adjust environmental controls (e.g., air conditioning, dehumidifiers) as needed to maintain the specified temperature and humidity levels.	1	2	-	-
PC7. verify that essential utilities like compressed air and power supply are available and functioning properly	1	1	-	-
PC8. conduct a visual inspection of utility lines and connections for any signs of damage or leaks	1	1	-	-
PC9. test the functionality of compressed air pressure and power supply voltage according to established procedures	1	2	-	-
PC10. document the test environment conditions (temperature, humidity, etc.) and any adjustments made according to SOPs	1	1	-	1
<i>Set up & Calibrate Wafer Test Equipment</i>	15	18	-	4
PC11. analyze the type of telecom wafers to be tested	1	1	-	-
PC12. review test program requirements for specific functionalities to be evaluated	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. identify the appropriate test equipment model from available options based on wafer type & test needs and consult SOPs for the selected test equipment model	1	2	-	1
PC14. identify and gather required test accessories like test heads and probes suitable for wafer testing	1	2	-	-
PC15. install the accessories onto the designated ports or fixtures on the test equipment, following established procedures, ensuring proper connection and secure positioning of the test heads and probes for optimal test performance	1	1	-	-
PC16. configure relevant equipment parameters (e.g., voltage levels, signal frequencies), based on the specific test program	1	1	-	1
PC17. ensure parameter settings align with the requirements outlined in the test program for accurate testing of functionalities (e.g., communication channel bandwidth, data integrity)	1	1	-	-
PC18. conduct a visual inspection of the test heads and probes to ensure proper alignment and positioning for contacting the designated test points on the wafers	1	1	-	-
PC19. utilize calibration tools or procedures outlined in SOPs to verify the calibration of test heads and probes	1	1	-	-
PC20. perform necessary adjustments or report the issue for maintenance personnel to address the issue, if calibration is out of specification	1	2	-	1
PC21. initiate a system check function on the test equipment as per SOPs	1	1	-	-
PC22. verify that all components of the test equipment, including accessories, are functioning properly as a system before proceeding with actual testing	1	1	-	-
PC23. identify the relevant test parameters requiring calibration for the wafer test	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. perform calibration routines for each identified parameter according to established procedures	1	1	-	1
PC25. document calibration results, including dates, performed adjustments, and reference standards used, for traceability purposes	1	1	-	-
<i>Load Wafers into the Test Equipment Following Handling Protocols</i>	15	18	-	3
PC26. review SOPs for specific handling procedures for the type of wafer being tested	1	1	-	-
PC27. visually inspect the wafers for any physical damage or defects that might affect testing	1	1	-	-
PC28. clearly label the wafers with identification information according to established protocols for traceability throughout the testing process	1	2	-	1
PC29. select and utilize appropriate handling tools (e.g., vacuum tweezers, transport carriers) designed for safe handling of delicate wafers	1	1	-	-
PC30. refer to SOPs or the test equipment manual for the designated loading pattern specific to the type of wafer being tested	2	2	-	-
PC31. verify that all wafers are loaded according to the pattern to optimize testing efficiency and avoid potential issues during the testing process	2	2	-	-
PC32. double-check the orientation of each loaded wafer to ensure it aligns with the test head and probe configuration for proper contact with the test points	2	2	-	-
PC33. verify that all wafers are securely positioned within the carrier or fixture to prevent any movement or misalignment during testing	1	2	-	1
PC34. secure the wafer carrier or fixture within the test equipment using the designated locking mechanisms	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. implement established electrostatic discharge (ESD) control procedures throughout the wafer handling process	2	2	-	1
PC36. utilize ESD-safe tools (e.g, grounded tweezers), wear ESD wrist straps grounded to the equipment, and maintain a clean and ESD-safe work environment	1	1	-	-
NOS Total	40	50	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7201
NOS Name	Assemble Wafer Test Equipment
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2025
NSQC Clearance Date	27/08/2024

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TEL/N7202: Perform Wafer Test Procedures

Description

This OS unit covers executing test programs, monitoring progress, and documenting results to evaluate wafer functionality.

Scope

The scope covers the following :

- Select & Execute Appropriate Test Program
- Monitor Test Progress & Document Test Results

Elements and Performance Criteria

Select & Execute Appropriate Test Program

To be competent, the user/individual on the job must be able to:

- PC1.** analyze the traveler sheets or batch records to determine the type of wafers being tested
- PC2.** review the specifications of the wafers, considering factors like their intended functionalities within a telecom system
- PC3.** consult the traveler sheets or batch records to understand the specific test requirements for the wafers
- PC4.** access the library of available test programs for the test equipment being used
- PC5.** select the appropriate test program that aligns with the specified test requirements, based on the identified wafer type and functionalities
- PC6.** review the documentation associated with the selected test program to understand its objectives
- PC7.** identify the scope of the test program, considering the specific functionalities it will assess within the wafer
- PC8.** initiate the process of loading the selected test program by following the SOPs (Standard Operating Procedures) for the test equipment
- PC9.** access the program settings within the loaded test program
- PC10.** configure relevant program parameters based on the specific test requirements, to ensure the test aligns with the intended evaluation
- PC11.** initiate the test program execution on the test equipment
- PC12.** monitor the test progress through the equipment interface, observing any real-time data displays or progress indicators
- PC13.** prepare to respond to prompts or error messages that might appear on the equipment interface during test execution
- PC14.** follow established procedures for handling prompts (e.g., user confirmation for specific test steps) and troubleshooting error messages (e.g., consulting troubleshooting guides or notifying maintenance personnel)

Monitor Test Progress & Document Test Results

To be competent, the user/individual on the job must be able to:

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- PC15.** analyze the real-time test data displayed on the equipment interface during test execution
- PC16.** interpret the meaning of the displayed data based on your understanding of the test program and the functionalities under evaluation
- PC17.** identify any trends, outliers, or irregularities that deviate from expected values or established pass/fail criteria, while monitoring the test data
- PC18.** configure suitable alarms and notifications within the test program or on the equipment itself to alert you of critical events during testing
- PC19.** analyze the test logs generated by the test program - detailed information about the test execution, including data points, measurements, and any error messages encountered
- PC20.** identify the potential root cause of any anomalies or failures observed during testing based on the analysis of real-time data and test logs
- PC21.** review the overall test outcomes, including pass/fail status for the wafers based on the defined criteria after test completion
- PC22.** utilize appropriate forms or electronic documentation tools as per company policy to document the test results
- PC23.** verify the accuracy and completeness of all documented test data, including parameters, measurements, and observations
- PC24.** identify passing and failing wafers, based on the defined pass/fail criteria established for the specific test program and wafer functionalities
- PC25.** document the concerns in the test records for any wafers exhibiting anomalies or borderline results that don't meet a clear pass/fail criterion
- PC26.** include details about the observed irregularities and any potential root causes identified during analysis
- PC27.** store the test data in a secure and accessible manner by following established company policies
- PC28.** ensure proper archiving procedures are followed to maintain the data for future reference and traceability purposes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of wafer tests conducted in semiconductor manufacturing (e.g., functional tests, parametric tests)
- KU2.** purpose and scope of various test programs used for different wafer types
- KU3.** interpretation of basic test data parameters (e.g., voltage, current, timing) and their significance
- KU4.** correlation between test results and potential wafer defects
- KU5.** selection criteria for choosing appropriate test programs based on wafer specifications
- KU6.** test program structure and configuration parameters
- KU7.** procedures for loading, initiating, and monitoring test program execution
- KU8.** troubleshooting techniques for common issues encountered during test execution
- KU9.** Statistical process control (SPC) techniques for analyzing test data trends
- KU10.** ability to identify outliers and anomalies in test data that may indicate potential problems

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- KU11.** B defect analysis methodologies for interpreting test results
- KU12.** procedures for escalating critical issues or anomalies requiring further investigation
- KU13.** data collection procedures and tools used to capture test results and parameters
- KU14.** standard operating procedures (SOPs) for documenting test data in designated forms or electronic systems
- KU15.** criteria for classifying wafers as pass or fail based on predefined test limits
- KU16.** archiving procedures for storing test data electronically or in paper format as per company policy
- KU17.** familiarity with the user interface of the wafer test equipment
- KU18.** functionalities and features available for test program execution
- KU19.** interpretation of real-time test data displayed on the equipment interface
- KU20.** procedures for responding to prompts, alarms, or error messages during testing

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work related records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the information/instructions being shared
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify appropriate solutions to work related issues
- GS7.** take quick decisions in case of an emergency/accident

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Select & Execute Appropriate Test Program</i>	19	31	-	5
PC1. analyze the traveler sheets or batch records to determine the type of wafers being tested	1	2	-	-
PC2. review the specifications of the wafers, considering factors like their intended functionalities within a telecom system	2	2	-	1
PC3. consult the traveler sheets or batch records to understand the specific test requirements for the wafers	1	2	-	-
PC4. access the library of available test programs for the test equipment being used	1	2	-	-
PC5. select the appropriate test program that aligns with the specified test requirements, based on the identified wafer type and functionalities	2	2	-	-
PC6. review the documentation associated with the selected test program to understand its objectives	1	1	-	-
PC7. identify the scope of the test program, considering the specific functionalities it will assess within the wafer	1	2	-	1
PC8. initiate the process of loading the selected test program by following the SOPs (Standard Operating Procedures) for the test equipment	2	3	-	-
PC9. access the program settings within the loaded test program	2	2	-	-
PC10. configure relevant program parameters based on the specific test requirements, to ensure the test aligns with the intended evaluation	2	3	-	1
PC11. initiate the test program execution on the test equipment	1	3	-	1
PC12. monitor the test progress through the equipment interface, observing any real-time data displays or progress indicators	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. prepare to respond to prompts or error messages that might appear on the equipment interface during test execution	1	2	-	-
PC14. follow established procedures for handling prompts (e.g., user confirmation for specific test steps) and troubleshooting error messages (e.g., consulting troubleshooting guides or notifying maintenance personnel)	1	3	-	1
<i>Monitor Test Progress & Document Test Results</i>	16	24	-	5
PC15. analyze the real-time test data displayed on the equipment interface during test execution	2	2	-	-
PC16. interpret the meaning of the displayed data based on your understanding of the test program and the functionalities under evaluation	1	2	-	-
PC17. identify any trends, outliers, or irregularities that deviate from expected values or established pass/fail criteria, while monitoring the test data	1	2	-	-
PC18. configure suitable alarms and notifications within the test program or on the equipment itself to alert you of critical events during testing	1	2	-	1
PC19. analyze the test logs generated by the test program - detailed information about the test execution, including data points, measurements, and any error messages encountered	1	1	-	-
PC20. identify the potential root cause of any anomalies or failures observed during testing based on the analysis of real-time data and test logs	2	2	-	1
PC21. review the overall test outcomes, including pass/fail status for the wafers based on the defined criteria after test completion	1	2	-	-
PC22. utilize appropriate forms or electronic documentation tools as per company policy to document the test results	1	1	-	-
PC23. verify the accuracy and completeness of all documented test data, including parameters, measurements, and observations	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. identify passing and failing wafers, based on the defined pass/fail criteria established for the specific test program and wafer functionalities	1	2	-	1
PC25. document the concerns in the test records for any wafers exhibiting anomalies or borderline results that don't meet a clear pass/fail criterion	1	1	-	-
PC26. include details about the observed irregularities and any potential root causes identified during analysis	1	2	-	-
PC27. store the test data in a secure and accessible manner by following established company policies	1	1	-	-
PC28. ensure proper archiving procedures are followed to maintain the data for future reference and traceability purposes	1	2	-	1
NOS Total	35	55	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7202
NOS Name	Perform Wafer Test Procedures
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2025
NSQC Clearance Date	27/08/2024

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TEL/N7203: Analyze Wafer Test Data

Description

This OS unit focuses on analyzing test data to identify trends, failing devices, and classifying wafers for further action.

Scope

The scope covers the following :

- Analyze & Evaluate Test Results
- Document, Report & Communicate Results

Elements and Performance Criteria

Analyze & Evaluate Test Results

To be competent, the user/individual on the job must be able to:

- PC1.** utilize statistical process control (SPC) techniques to analyze the collected test data for wafers
- PC2.** select appropriate SPC charts based on the specific test parameters and wafer functionalities
- PC3.** interpret control chart data to identify trends, shifts, or outliers that might indicate potential process issues
- PC4.** calculate statistical metrics to assess process capability and identify opportunities for improvement
- PC5.** compare the test data for each wafer against established criteria or historical data, considering tolerances and expected variations
- PC6.** identify any outliers, unexpected dips or spikes in parameters, or deviations from normal test results, categorizing them by severity (critical, minor)
- PC7.** utilize wafer maps or reference markers to correlate specific test data points with their corresponding locations on the wafer
- PC8.** analyze spatial patterns of defects to identify potential causes related to specific locations within the wafer fabrication process
- PC9.** apply knowledge of circuit design principles and Semiconductor Manufacturing Processes to understand the potential causes of the identified defects based on the type of test performed
- PC10.** consult with design engineers or process specialists to gain further insights into potential root causes based on their expertise
- PC11.** classify failing devices based on the type of defect (e.g., electrical, physical) and its severity level (critical, moderate, marginal)
- PC12.** evaluate the impact of failing devices on overall wafer functionality, considering redundancy or isolation mechanisms within the circuit design
- PC13.** recommend appropriate next steps for failing devices, including retesting, scrapping, or further investigation using specialized equipment
- PC14.** analyze data from multiple wafers and test runs to distinguish between random defects (isolated incidents) and systematic problems (affecting multiple devices or wafers)

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- PC15.** utilize statistical analysis tools to identify patterns or trends that suggest systematic issues within the manufacturing process
- PC16.** recommend corrective actions or process improvements to address identified systematic problems and prevent future occurrences

Document, Report & Communicate Results

To be competent, the user/individual on the job must be able to:

- PC17.** apply the pre-defined pass/fail criteria established for the specific wafer test program to categorize each wafer
- PC18.** document the classification (pass/fail) of each wafer and any associated defect information in the designated test record or electronic documentation system
- PC19.** utilize data visualization tools to represent test results and highlight key findings for improved clarity in reports
- PC20.** archive test data and associated documentation securely and electronically as per company policy for future reference and traceability purposes
- PC21.** compile well-structured and concise test reports summarizing the findings from the wafer testing process
- PC22.** include key information like overall pass/fail rates, defect classifications with severity levels, and any identified trends or anomalies
- PC23.** tailor test reports to the audience (e.g., engineers, quality control) by emphasizing relevant details and technical aspects
- PC24.** disseminate the test data and findings to designated personnel through meetings, emails, or a centralized information sharing platform, following established communication protocols

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles and techniques of SPC for analyzing wafer test data.
- KU2.** identifying trends, patterns, and outliers in test data using SPC charts.
- KU3.** interpreting process capability based on statistical analysis.
- KU4.** correlation between test data parameters and potential wafer defects.
- KU5.** techniques for analyzing and interpreting complex test data sets.
- KU6.** identifying failing device locations on the wafer based on test data
- KU7.** basic concepts of semiconductor device failure mechanisms.
- KU8.** techniques for root cause analysis of failing devices based on test data.
- KU9.** differentiating between random defects and systematic issues.
- KU10.** defined criteria and test limits for classifying wafers as pass or fail.
- KU11.** impact of failing devices on overall wafer yield and functionality.
- KU12.** procedures for marking and flagging failing wafers for further investigation.
- KU13.** techniques for preparing clear and concise reports summarizing test data.
- KU14.** importance of highlighting critical anomalies and failing devices.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** maintain work related records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the information/instructions being shared
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify appropriate solutions to work related issues
- GS7.** take quick decisions in case of an emergency/accident

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyze & Evaluate Test Results</i>	25	35	-	5
PC1. utilize statistical process control (SPC) techniques to analyze the collected test data for wafers	2	2	-	1
PC2. select appropriate SPC charts based on the specific test parameters and wafer functionalities	1	2	-	-
PC3. interpret control chart data to identify trends, shifts, or outliers that might indicate potential process issues	1	2	-	-
PC4. calculate statistical metrics to assess process capability and identify opportunities for improvement	2	2	-	-
PC5. compare the test data for each wafer against established criteria or historical data, considering tolerances and expected variations	1	2	-	-
PC6. identify any outliers, unexpected dips or spikes in parameters, or deviations from normal test results, categorizing them by severity (critical, minor)	2	2	-	1
PC7. utilize wafer maps or reference markers to correlate specific test data points with their corresponding locations on the wafer	1	2	-	1
PC8. analyze spatial patterns of defects to identify potential causes related to specific locations within the wafer fabrication process	2	3	-	1
PC9. apply knowledge of circuit design principles and Semiconductor Manufacturing Processes to understand the potential causes of the identified defects based on the type of test performed	2	3	-	1
PC10. consult with design engineers or process specialists to gain further insights into potential root causes based on their expertise	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. classify failing devices based on the type of defect (e.g., electrical, physical) and its severity level (critical, moderate, marginal)	1	2	-	-
PC12. evaluate the impact of failing devices on overall wafer functionality, considering redundancy or isolation mechanisms within the circuit design	2	2	-	-
PC13. recommend appropriate next steps for failing devices, including retesting, scrapping, or further investigation using specialized equipment	2	2	-	-
PC14. analyze data from multiple wafers and test runs to distinguish between random defects (isolated incidents) and systematic problems (affecting multiple devices or wafers)	1	2	-	-
PC15. utilize statistical analysis tools to identify patterns or trends that suggest systematic issues within the manufacturing process	2	2	-	-
PC16. recommend corrective actions or process improvements to address identified systematic problems and prevent future occurrences	1	2	-	-
<i>Document, Report & Communicate Results</i>	10	20	-	5
PC17. apply the pre-defined pass/fail criteria established for the specific wafer test program to categorize each wafer	1	3	-	1
PC18. document the classification (pass/fail) of each wafer and any associated defect information in the designated test record or electronic documentation system	1	2	-	1
PC19. utilize data visualization tools to represent test results and highlight key findings for improved clarity in reports	1	3	-	-
PC20. archive test data and associated documentation securely and electronically as per company policy for future reference and traceability purposes	1	2	-	1
PC21. compile well-structured and concise test reports summarizing the findings from the wafer testing process	1	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. include key information like overall pass/fail rates, defect classifications with severity levels, and any identified trends or anomalies	2	2	-	-
PC23. tailor test reports to the audience (e.g., engineers, quality control) by emphasizing relevant details and technical aspects	2	3	-	1
PC24. disseminate the test data and findings to designated personnel through meetings, emails, or a centralized information sharing platform, following established communication protocols	1	2	-	1
NOS Total	35	55	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7203
NOS Name	Analyze Wafer Test Data
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2025
NSQC Clearance Date	27/08/2024

Qualification Pack

TEL/N7204: Maintain Wafer Test Equipment

Description

This OS unit covers performing routine maintenance, troubleshooting, and reporting issues to ensure optimal wafer test equipment performance.

Scope

The scope covers the following :

- Perform Routine Cleaning and Preventive Maintenance of Equipment
- Identify, Troubleshoot & Report Equipment Malfunctions

Elements and Performance Criteria

Perform Routine Cleaning and Preventive Maintenance of Equipment

To be competent, the user/individual on the job must be able to:

- PC1.** follow established SOPs (Standard Operating Procedures) for cleaning various components of the wafer test equipment
- PC2.** identify and utilize appropriate cleaning materials and equipment for specific components
- PC3.** perform cleaning tasks to ensure a dust-free and contamination-free environment within the test equipment, minimizing potential issues like test signal degradation or device damage
- PC4.** ensure cleaning of areas crucial for proper test execution (e.g., test fixtures, probe tips, wafer contact surfaces)
- PC5.** identify and perform routine maintenance activities on the test equipment according to established schedules (eg. lubricating moving parts, replacing air filters, and calibrating specific test instruments to ensure optimal performance)
- PC6.** maintain a log of completed preventive maintenance activities for each piece of equipment, including dates and details of actions performed
- PC7.** visually inspect the equipment for any signs of wear or damage during cleaning and maintenance procedures, reporting any concerns to appropriate personnel
- PC8.** organize and maintain a clean and clutter-free workspace around the test equipment to prevent accidental damage or contamination of wafers and test components
- PC9.** properly dispose of used cleaning materials and waste generated during maintenance activities in designated bins or containers
- PC10.** keep tools and equipment used for cleaning and maintenance organized and readily accessible for efficient use
- PC11.** report any spills or contamination incidents within the work area promptly for proper cleaning and containment

Identify, Troubleshoot & Report Equipment Malfunctions

To be competent, the user/individual on the job must be able to:

- PC12.** recognize potential dangers and indications of equipment failure during operation, such as unusual noises, error messages, or unexpected behavior in test results

Qualification Pack

- PC13.** utilize equipment manuals, troubleshooting guides, and diagnostic tools (e.g., built-in self-test functions) to identify the source of potential failures
- PC14.** observe and document specific details about the malfunction, including error messages displayed, test interruptions, or abnormal equipment behavior
- PC15.** consult with experienced personnel (e.g., engineers, maintenance technicians) for further diagnosis and guidance, when encountering unfamiliar issues
- PC16.** perform basic troubleshooting steps as per established procedures or manufacturer recommendations, such as restarting the equipment, checking cable connections, or resetting specific settings
- PC17.** document the troubleshooting steps taken and their outcomes for future reference and to inform further repair efforts
- PC18.** clearly identify the problem and its impact on test operations (e.g., complete test stoppage, partial functionality loss), if basic troubleshooting fails to resolve the issue
- PC19.** report equipment malfunctions requiring repair work beyond basic troubleshooting capabilities to designated personnel (e.g., maintenance technicians) following established company protocols
- PC20.** record all maintenance activities performed on the test equipment in a detailed and comprehensive manner, including cleaning, preventive maintenance, and repairs
- PC21.** utilize designated logbooks or a computerized maintenance management system to document maintenance activities, including the date, type of maintenance performed, specific actions taken, replacement parts used (if applicable), and any observations made
- PC22.** adhere to company policies for organizing and storing maintenance logs and records electronically or in designated physical binders for future reference and traceability purposes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of preventive maintenance for maintaining optimal equipment performance
- KU2.** basic principles of cleaning procedures to prevent contamination within the equipment
- KU3.** procedures for performing routine maintenance tasks as outlined in SOPs (Standard Operating Procedures)
- KU4.** safe handling and storage procedures for cleaning materials and tools
- KU5.** common malfunctions encountered during wafer test equipment operation
- KU6.** techniques for identifying the root cause of equipment issues using troubleshooting flowcharts
- KU7.** limitations of their own troubleshooting abilities and when to escalate issues to qualified personnel
- KU8.** importance of clear and concise communication when reporting equipment malfunctions
- KU9.** procedures for documenting cleaning, maintenance, and repair activities in designated logs or electronic systems
- KU10.** the importance of maintaining accurate and complete maintenance records for future reference
- KU11.** company policies for organizing and storing maintenance logs and records for traceability

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- KU12.** utilizing preventive maintenance schedules to identify upcoming maintenance tasks based on documented history
- KU13.** established procedures for reporting equipment malfunctions to designated personnel (e.g., maintenance technicians, engineers)
- KU14.** importance of clear and concise communication when reporting issues, including error messages and observations
- KU15.** the role of maintenance logs and records in facilitating communication with maintenance personnel

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work related records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the information/instructions being shared
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify appropriate solutions to work related issues
- GS7.** take quick decisions in case of an emergency/accident

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform Routine Cleaning and Preventive Maintenance of Equipment</i>	12	24	-	4
PC1. follow established SOPs (Standard Operating Procedures) for cleaning various components of the wafer test equipment	1	3	-	1
PC2. identify and utilize appropriate cleaning materials and equipment for specific components	1	3	-	1
PC3. perform cleaning tasks to ensure a dust-free and contamination-free environment within the test equipment, minimizing potential issues like test signal degradation or device damage	1	2	-	-
PC4. ensure cleaning of areas crucial for proper test execution (e.g., test fixtures, probe tips, wafer contact surfaces)	1	2	-	-
PC5. identify and perform routine maintenance activities on the test equipment according to established schedules (eg. lubricating moving parts, replacing air filters, and calibrating specific test instruments to ensure optimal performance)	2	3	-	-
PC6. maintain a log of completed preventive maintenance activities for each piece of equipment, including dates and details of actions performed	1	1	-	-
PC7. visually inspect the equipment for any signs of wear or damage during cleaning and maintenance procedures, reporting any concerns to appropriate personnel	1	2	-	-
PC8. organize and maintain a clean and clutter-free workspace around the test equipment to prevent accidental damage or contamination of wafers and test components	1	2	-	-
PC9. properly dispose of used cleaning materials and waste generated during maintenance activities in designated bins or containers	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. keep tools and equipment used for cleaning and maintenance organized and readily accessible for efficient use	1	3	-	1
PC11. report any spills or contamination incidents within the work area promptly for proper cleaning and containment	1	2	-	1
<i>Identify, Troubleshoot & Report Equipment Malfunctions</i>	18	36	-	6
PC12. recognize potential dangers and indications of equipment failure during operation, such as unusual noises, error messages, or unexpected behavior in test results	2	4	-	1
PC13. utilize equipment manuals, troubleshooting guides, and diagnostic tools (e.g., built-in self-test functions) to identify the source of potential failures	2	3	-	1
PC14. observe and document specific details about the malfunction, including error messages displayed, test interruptions, or abnormal equipment behavior	1	3	-	1
PC15. consult with experienced personnel (e.g., engineers, maintenance technicians) for further diagnosis and guidance, when encountering unfamiliar issues	2	3	-	1
PC16. perform basic troubleshooting steps as per established procedures or manufacturer recommendations, such as restarting the equipment, checking cable connections, or resetting specific settings	2	4	-	-
PC17. document the troubleshooting steps taken and their outcomes for future reference and to inform further repair efforts	2	3	-	1
PC18. clearly identify the problem and its impact on test operations (e.g., complete test stoppage, partial functionality loss), if basic troubleshooting fails to resolve the issue	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. report equipment malfunctions requiring repair work beyond basic troubleshooting capabilities to designated personnel (e.g., maintenance technicians) following established company protocols	1	4	-	-
PC20. record all maintenance activities performed on the test equipment in a detailed and comprehensive manner, including cleaning, preventive maintenance, and repairs	1	3	-	-
PC21. utilize designated logbooks or a computerized maintenance management system to document maintenance activities, including the date, type of maintenance performed, specific actions taken, replacement parts used (if applicable), and any observations made	1	3	-	-
PC22. adhere to company policies for organizing and storing maintenance logs and records electronically or in designated physical binders for future reference and traceability purposes	2	3	-	1
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7204
NOS Name	Maintain Wafer Test Equipment
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2025
NSQC Clearance Date	27/08/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

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5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70% aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N7201.Assemble Wafer Test Equipment	40	50	-	10	100	20
TEL/N7202.Perform Wafer Test Procedures	35	55	-	10	100	25
TEL/N7203.Analyze Wafer Test Data	35	55	-	10	100	25
TEL/N7204.Maintain Wafer Test Equipment	30	60	-	10	100	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	160	250	-	40	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.